VYOMAA LABS

AI-Enabled Unified First Information Report (u-FIR)

Technical Proposal

Submitted to: Andhra Pradesh Prohibition & Excise Department
via AI Living Labs Hackathon Arena

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Website: https://vyomaalabs.github.io/vyomaa-labs/

1. Executive Summary

The Excise Department generates thousands of case records a year — FIRs, investigation diaries, progress reports, chargesheets, court documents — all written as free-flowing text, with no two stations recording the same details the same way, and the same offender's name spelled differently across files. The result: years of institutional knowledge that no officer can actually search, compare, or learn from at scale.

VYOMAA LABS proposes a system that reads the department's case records exactly as they already exist — no new forms, no change to how officers write — and turns them into structured, searchable, queryable intelligence. It suggests applicable legal sections instead of relying on an officer's memory, resolves name and location variants to surface repeat offenders and criminal networks, drafts case diaries and chargesheets for officer review, and answers natural-language questions through a role-based, Telugu-capable chatbot.

The proposal is built to be judged, not just read: every section below is written to map directly onto the hackathon's own scoring rubric (Section 13 gives the explicit alignment table), and the architecture is designed for the only test that matters for a law-enforcement system — that it keeps working, keeps a complete audit trail, and never makes a filing decision without an officer's sign-off.

2. Problem Understanding

Restated in our own words, to show precisely what we are building against:

- Case data is unstructured. Who was involved, what was seized, which laws applied — all of it sits buried inside long narrative text rather than structured fields.
- Records are inconsistent across stations, and the same person or place can appear under multiple spellings across different case files — a classic entity-resolution problem, not a data-entry one.
- Institutional memory is personal, not organizational. Spotting a repeat offender or a network today depends on an individual officer happening to remember a name from a previous case.
- The department explicitly does not want officers to change how they write. The system has to work on existing free text, not require new structured intake forms.

The department has stated five concrete goals: digitize and structure all case records; auto-generate case diaries and chargesheets; auto-detect suspect and network links; identify crime hotspots from the data; and provide a role-based chatbot for officers. The brief also specifies the data on offer — five years of crime case records, including detected violations, decoy operation results, and repeat-offender history — and requires the solution to expose results via an API or dashboard. The pilot that follows a successful hackathon result is concrete too: three Excise districts, five to ten department users, sixty days, and a target of over 90% accuracy in predicting risk. Every part of the design below is sized to that pilot, not to an abstract future scale.

3. Proposed Solution Overview

The system is a pipeline of specialized AI capabilities coordinated by an orchestrating agent, not a single monolithic model asked to do everything at once. Each stage has one job, a defined input/output contract, and its own fallback behavior — this is what makes the system auditable, debuggable, and resilient to any one component failing, rather than a black box that either works or doesn't.

- Ingestion & OCR — accepts typed records directly, and scanned or handwritten pages via OCR/handwriting recognition, normalizing everything to a common text representation.
- Structuring (NER) — extracts people, places, vehicles, seized items, dates, and offense details from the free text into structured fields, without requiring the source text to change.
- Legal Section Advisor — a retrieval-augmented component that suggests applicable sections of the Excise Act and relevant IPC/BNS provisions based on the facts extracted, citing the specific statute text it relied on so an officer can verify the reasoning, not just trust a label.
- Entity Resolution — matches name and location variants across records ("J. Reddy" / "Jagan Reddy") using a layered approach: deterministic ID matches first, embedding-based similarity for likely aliases, and an LLM reconciliation pass with a confidence score for ambiguous cases, with low-confidence merges routed to human review rather than applied automatically.
- Network & Hotspot Graph — builds a knowledge graph of resolved people, places, vehicles, and cases to surface repeat offenders, likely networks, and geographic hotspots through graph traversal and link analysis.
- Case Diary / Chargesheet Drafting — produces a structured first draft from the extracted facts for the investigating officer to edit and approve. The system never auto-files a legal document; it drafts, the officer signs.
- Role-Based Chatbot — a natural-language query interface over the structured data and graph, available in English and Telugu via Bhashini integration, scoped to what each officer's role is permitted to see.

This mirrors the design pattern already proven at scale in ambient clinical documentation tools such as AWS HealthScribe, which turns unstructured doctor-patient conversation into structured clinical notes with every generated line traceable back to its source — the same trace-to-source principle applies here: every suggested legal section, every flagged network link, and every chatbot answer must cite the source record it came from.

4. System Architecture

4.1 Orchestration pattern

The system follows a router-plus-specialist-agents pattern now standard in production agentic systems: an orchestrator agent receives a request (a new case record, or an officer's question), decides which specialist stage(s) it needs, calls them as tools with structured inputs/outputs, and assembles the result. Simple lookups ("show me case #4521") route directly to the data layer without invoking heavier reasoning stages; complex queries ("find anyone connected to this vehicle in the last three years") route through the graph-traversal and link-analysis stages. This keeps routine queries fast and cheap while reserving the expensive reasoning path for where it's actually needed.

4.2 Data flow

- Case record (typed or scanned) enters via the ingestion API.
- Orchestrator routes it through OCR (if needed) → structuring → legal section advisor → entity resolution → graph upsert, each stage writing its output to an auditable event log before the next stage reads it.
- Resolved entities and relationships are merged into the knowledge graph using upsert semantics, so reprocessing a record never creates duplicate nodes.
- The raw extracted text remains in a vector store alongside the graph — the graph is a reasoning aid built on top of the data, never the system of record itself.

- Dashboard and API expose structured records, network graphs, hotspot maps, and chatbot access, scoped by officer role.

4.3 Entity resolution — the component most of these systems get wrong

Research into production knowledge-graph systems is consistent on one point: entity resolution, not model quality, is where these systems typically fail — two spellings of one name silently becoming two people in the graph, which then makes every downstream network-detection query wrong without any visible error. We address this directly with a layered matching strategy (deterministic rules, then embedding similarity, then LLM reconciliation with confidence scoring), human review for any low-confidence or high-impact merge, and an audit trail on every merge decision so a contested link can always be traced back to why the system made it.

5. Technology Stack

Layer	Approach
Ingestion / OCR	Document & handwriting OCR pipeline; typed-text path bypasses OCR entirely
Structuring / NER	Fine-tuned extraction model for case-domain entities (people, places, vehicles, items, offenses)
Legal retrieval	RAG over a curated, versioned corpus of Excise Act / IPC / BNS sections, with citation of source text
Entity resolution	Deterministic rules + embedding similarity + LLM reconciliation, each merge confidence-scored
Knowledge graph	Graph database for repeat-offender and network link analysis, with upsert-based updates
Orchestration	Router-agent pattern: an orchestrator calls specialist stages as tools, each independently testable
Primary inference	Managed LLM API for heavier reasoning stages (drafting, chat), behind a provider-agnostic interface
Fallback inference	Self-hosted, on-premise open-weight model (see Section 6) as a guaranteed-available fallback
Interface	Role-based web dashboard, REST/GraphQL API (per the brief's requirement), and a Bhashini-integrated chatbot
Data residency	Deployed within government infrastructure (state data center / MeghRaj); no case data leaves sovereign infrastructure

6. Reliability, Scalability & Failure Handling

This is designed as production government infrastructure, not a demo. Every stage in Section 4 can fail independently without taking the rest of the system down, and every failure is recoverable rather than silent.

6.1 No single point of failure

- Each pipeline stage is an independently deployed, independently scaled service, connected by an async message queue (not direct synchronous calls) — if the legal-section advisor is briefly unavailable, ingestion and structuring continue, and the queued record is processed as soon as it recovers.

- Failed stage invocations retry automatically with backoff; anything that still fails after retries lands in a dead-letter queue for inspection and reprocessing, rather than silently dropping the record.
- A circuit breaker sits in front of every external AI API call: if the primary managed-LLM provider is slow or unavailable, traffic automatically falls back to the self-hosted open-weight model (the same class of model — Qwen-class small language models — VYOMAA LABS already runs fully self-hosted today) rather than the system going down.

6.2 Data durability

- Automated, encrypted, geo-redundant backups on a 3-2-1 basis (three copies, two media types, one off-site within sovereign infrastructure), with point-in-time recovery.
- Every AI-assisted action — every suggested section, every entity merge, every chatbot answer — is written to an append-only, tamper-evident audit log. This directly satisfies the brief's own requirement that "every AI-assisted decision is recorded and can be reviewed later."

6.3 Scaling from pilot to statewide

The pilot scope (3 districts, 5–10 users, 60 days) runs comfortably on a single regional deployment. The same architecture scales horizontally for statewide rollout: stateless services scale out under load, the message-queue layer absorbs bursty ingestion (e.g., bulk-loading five years of historical records) without overwhelming downstream stages, and the knowledge graph shards by district while supporting cross-district queries for state-level network detection — the exact capability the brief asks for ("comparing cases across the state").

7. Data Privacy & Security

Data sovereignty is the organizing principle, not an afterthought. VYOMAA LABS has already demonstrated — on our own hardware, not in a slide — that we can run real language-model inference fully self-hosted with zero dependency on any foreign cloud AI API. That same approach scales directly into an on-premise, air-gapped-capable deployment inside government infrastructure (state data center or MeghRaj), which matters specifically because this is law-enforcement data: case records, informant and decoy-operation details, and repeat-offender histories that should never have to leave sovereign infrastructure to be processed.

- Role-based access control, scoped by officer rank, station, and case assignment — no officer sees more than their role requires.
- Encryption at rest (AES-256) and in transit (TLS) for all case data, at every stage of the pipeline.
- Immutable, append-only audit logging of every AI-assisted decision, satisfying the brief's review requirement and giving the department a complete, tamper-evident record for any legal challenge to an AI-suggested section or network link.
- No case data is used to train or fine-tune any third-party model; any model fine-tuning happens on self-hosted infrastructure, on data that never leaves government control.

8. Accuracy & Validation

The system is built around one hard rule: it drafts and suggests, officers decide and sign. No case diary, chargesheet, legal-section suggestion, or network link is ever auto-filed or treated as a final determination without

a human officer's explicit approval. This is not a limitation we're apologizing for — it is the correct design for a law-enforcement AI system, and it is also what keeps the system legally defensible.

- Every suggestion carries a confidence score and a citation back to the source text it was derived from, so an officer can verify, not just trust.
- Accuracy is tracked continuously against the pilot's own 90% target, broken down by stage (extraction accuracy, section-suggestion precision, entity-merge precision/recall) rather than one blended number, so any weak stage is visible and fixable on its own.
- We take the lesson of the COMPAS recidivism-scoring controversy seriously: that system's core failure was not technical inaccuracy but an undisclosed, proprietary risk score used directly in sentencing decisions with no way for the affected person to inspect or contest it, and a fairness definition nobody had explicitly agreed on. We avoid every one of those failure modes by design — every score is explainable and source-cited, nothing is proprietary-black-box, nothing is used for an automated decision, and fairness criteria (which error types matter most, across which groups) are defined explicitly and reviewably rather than left implicit in the training data.
- Bias auditing is a standing process, not a one-time check: extraction and section-suggestion accuracy is periodically reviewed across stations and case types to catch any systematic skew before it compounds.

9. Usability & Accessibility

The brief is explicit that officers should not have to change how they write — so the only new thing any officer interacts with is the dashboard, API, and chatbot, not a new intake form.

- Role-based dashboard: a station-level officer, a district SP, and a state-level analyst see different views of the same underlying data, scoped to what's relevant and permitted for their role.
- Bhashini-integrated chatbot: the department's own hackathon platform already uses Bhashini (India's national multilingual AI initiative) for translation — we extend that same integration so officers can query the system in Telugu as naturally as in English, rather than forcing English-only interaction onto a predominantly Telugu-speaking workforce.
- API access for integration with existing systems (CCTNS and, in time, ICJS), rather than becoming one more disconnected silo — the brief's "share results through an API or dashboard" requirement is treated as a first-class design constraint, not an afterthought.
- Every AI output (a suggested section, a flagged link) shows its reasoning and source in plain language, not just a score, so adoption doesn't depend on officers blindly trusting a black box.

10. Impact Potential

- Officer time shifts from paperwork back to investigation — the brief's own stated goal.
- Legal section suggestions become consistent across stations and districts, rather than depending on which officer happens to be on the case.
- Hidden criminal networks and repeat offenders become visible at a state level for the first time, rather than living only in individual officers' memory.
- Five years of historical case data — currently an unused archive — becomes an active resource: searchable, comparable, and usable for pattern detection across the whole state.
- A successful pilot creates a template directly reusable by other Andhra Pradesh departments sitting on the same kind of unstructured records, and by CCTNS/ICJS integration efforts more broadly.

11. Phase Breakdown / Roadmap

Phase	Window	Deliverable
Application	By 30 Oct 2026	This proposal + live website submitted; architecture and pilot plan finalized
Shortlist prep	30 Oct – 3 Nov 2026	Working prototype on sample case data: ingestion → structuring → section suggestion demonstrated end-to-end
Presentation	By 13 Nov 2026	Live demo for the jury: a real (anonymized) case record processed through the full pipeline, including the chatbot
Decision	16 Nov 2026	Winner announcement
Pilot — setup	Weeks 1–2 of pilot	Deployed in 3 selected Excise districts; 5–10 department users onboarded; historical 5-year data ingested
Pilot — operation	Weeks 3–8 of pilot (60 days total)	Live use on new cases; weekly accuracy tracking against the 90% target, stage-by-stage
Pilot — review	End of 60 days	Accuracy report, officer feedback, go/no-go recommendation for expansion
Statewide rollout	Post-pilot	Phased expansion district by district, with CCTNS/ICJS integration as a standing workstream

12. Team

VYOMAA LABS is a DPIIT-recognized startup (Certificate No. DIPP290455, registered as a Partnership Firm) — a small, focused team building AI that does real work close to the people using it. Before writing this proposal, we had already built and run a small language model fully self-hosted end-to-end — not as a planned demo, but as working infrastructure — which is the direct, hands-on root of the data-sovereignty architecture proposed in Section 7. We build things and run them, not just design them on slides.

13. Criteria Alignment

Mapped directly to the hackathon's published scoring rubric, so each criterion can be checked against a specific section of this document.

Admin / screening criteria

Criterion	Weight	Addressed in
Innovation and AI Advancement	25%	Sections 3, 4 — multi-stage agentic pipeline, GraphRAG-based entity resolution and link analysis, self-hosted sovereign fallback
Accuracy and Reliability	20%	Sections 6, 8 — stage-level accuracy tracking against the 90% pilot target, human sign-off on every output
Usability and Accessibility	15%	Section 9 — role-based views, Bhashini Telugu integration, no change to officer writing habits
Scalability and Deployment	15%	Section 6.3 — horizontal scaling, district sharding, pilot-to-

		statewide path
Data Privacy and Security	10%	Section 7 — sovereign on-premise deployment, RBAC, encryption, immutable audit log
Impact Potential	15%	Section 10

Jury criteria

Criterion	Weight	Addressed in
Relevance to Problem Statement	15%	Section 2 — restated directly from the brief; Section 3 addresses each of the five stated goals by name
Innovation and Novelty	10%	Section 4.3 — layered entity resolution designed around the specific failure mode (name-spelling variants) the brief itself names
Technical Feasibility and Scalability	15%	Sections 5, 6
Data and Integration Capability	10%	Section 9 — API-first, built for CCTNS/ICJS integration rather than a silo
User Experience and Accessibility	20%	Section 9 — Bhashini multilingual support, role-based dashboard
Impact and Outcome Potential	20%	Section 10
Presentation and Demo Quality	10%	Live end-to-end demo planned for the 13 Nov presentation per Section 11

14. References

- Crime and Criminal Tracking Network and Systems (CCTNS) — Ministry of Home Affairs / NCRB, national backbone for FIR, investigation and chargesheet digitization, live in over 95% of India's police stations.
- Interoperable Criminal Justice System (ICJS) — MHA, links CCTNS, e-Courts, e-Prosecution, e-Forensics and e-Prisons on a shared data model; the integration target this proposal is built toward.
- HOLMES 2 (UK) — the design precedent for centralizing major-case information around a structured incident record rather than scattered free text.
- AWS HealthScribe / ambient clinical documentation — the direct architectural analogue for turning unstructured narrative (clinician-patient conversation; here, case narrative) into structured, source-cited records without changing how the source professional works.
- ProPublica's COMPAS investigation and the subsequent fairness-definition literature — the cautionary precedent this proposal is explicitly designed against: no proprietary black-box scoring, no automated decisions, explicit and reviewable fairness criteria.
- Current practitioner consensus on GraphRAG and agentic orchestration (2026) — entity resolution as the primary failure point in knowledge-graph systems, addressed here via layered matching with confidence scoring and human review, per Section 4.3.